

AMD Motherboards				
Brand Model No.	Palit AA-780G	XFX MI-A78S-8209	XFX MI-A78U-8309	ZOTAC GeForce 8300
				
Chipset	AMD 780G	NVIDIA GeForce 8200	NVIDIA GeForce 8300	NVIDIA GeForce 8300
Socket	AM2+	AM2+	AM2+	AM2+
Northbridge	AMD 780G	NVIDIA GeForce 8200	NVIDIA GeForce 8300	NVIDIA GeForce 8300
Southbridge	AMD SB700			
Price (Rs)	3,700	5,500	6,000	5,999
+	Good value for money	Decent performance	Decent performance	Above average performer
-	Lacks HDMI	Nothing in particular	Nothing in particular	Insufficient cooling
Grand Totals (Out Of 100)	50.79	57.28	58.77	57.22
Features (Out Of 80)	29.40	36.61	37.31	35.46
Performance (Out Of 20)	21.39	20.67	21.46	21.76
Features				
Memory Type / Max Freq. Supported	DDR2 / 1066 MHz	DDR2 / 1066 MHz	DDR2 / 1066 MHz	DDR2 / 1066 MHz
Memory Supported / No. Of slots	8 GB / 4	8 GB / 4	8 GB / 4	8 GB / 4
Power Saving features (with GPU)	✗	✓	✓	✓
No. Of SATA / PATA Ports	6 / 1	4 / 1	6 / 1	6 / 1
No. Of Rear E-SATA Ports	0	1	1	0
HDMI / DVI / VGA	✗ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✗ / ✓ / ✓
Multi GPU Support (SLI / CrossFire / N)	✗	✗	✗	✗
Audio Controller / Channels / HD Compliant (Y / N)	ALC883 / 8 / ✓	ALC889 / 8 / ✓	ALC889 / 8 / ✓	ALC888 / 8 / ✓
S / PDIF / Optical Port	✗ / ✗	✗ / ✗	✗ / ✗	✓ / ✗
No Of LAN Ports / 10 / 100 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit
Integrated WiFi (Y / N)	✗	✗	✗	✗
No. Of Rear USB / FireWire Ports	4 / 0	4 / 0	4 / 0	4 / 1
All Solid State Capacitors (Y / N)	✗	✗	✗	✗
Overclocking Options				
CPU / FSB / Memory	✗ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✗ / ✗
Overvolting (CPU / Memory / FSB / PCIe / Northbridge)	✗ / ✓ / ✗ / ✓ / ✗	✓ / ✓ / ✓ / ✗ / ✓	✓ / ✓ / ✓ / ✗ / ✓	✗ / ✓ / ✓ / ✗ / ✗
Package Bundle				
No of SATA / PATA Cables Provided	1 / 1	1 / 1	1 / 1	2 / 1
Key bundled items				
Performance				
PC Mark 2005				
CPU	7588	7610	7620	7626
Memory / Graphics	4288 / 2078	4499 / 2104	4509 / 2189	4780 / 2186
HDD	6040	6140	6143	6148
Overall	5401	5517	5548	5643
SiSoft Sandra 2008				
CPU Arithmetic (Dhrystone / Whetstone)	39339 / 33394	38898 / 32955	38999 / 33029	36588 / 33116
CPU Multimedia (Integer / Floating)	98660 / 130102	97192 / 128341	97828 / 129058	97815 / 128991
Drive Index (MBps) / Random Access Time (ms)	80 / 12	80 / 13	80 / 13	80 / 13
Memory Bandwidth Score (Integer / Floating)	5021 / 5023	4971 / 4978	5204 / 5197	5159 / 5167
3D Mark 2005:				
3D Marks / CPU Score	3052 / 5517	2347 / 8285	2529 / 8050	2556 / 8642
Video Encoding (DivX 6.2) Time	96	94	93	93
File Transfer - 4 GB File	199	213	200	195
Doom 3 (640x480 / 1024x768 - ultra no AA/AF)	48.8 / 28.4	57.7 / 30.8	59.7 / 32.6	60.3 / 32.9
Company of Heroes (800x600/1280x1024)	25.2 / 12.7	15.2 / 9.4	17.7 / 10.6	17.7 / 10.9

things are also missing. The biggest difference can be seen by simply looking at the board. There are no heatpipes and there is a very basic solution in the form of a heatsink. Everything else about the board is still great. The board has loads of space, three PCI slots but just no SLI. Even the capacitors are cleanly put in small bunches whenever necessary. SATA ports are mounted sideways, so the cables don't interfere with the card.

The two boards from Jetway had very similar designs and layout. The two based on the AMD

780G and the 790GX chipsets. The towering heatsinks on the boards aren't of the best quality and they don't seem mounted very well. The SATA ports lie in between the two PCI-E slots so they steer clear of any large cards that you might use on the board.

The MSI DKA790GX Platinum is a huge board and is absolutely unlike the G43M2 board for Intel processors. The heatsink on the chipset isn't very good though. The fins are delicate and easily break off if you aren't careful while fitting the processor heatsink.

The NVIDIA 8200 and 8300-based motherboards from Biostar, XFX, ZOTAC were all compact Micro ATX boards. The ZOTAC GeForce 8300 especially got very hot when we ran benchmarks. Overclocking with a chipset this hot is pretty unthinkable. The Biostar GF8200 M2+ had an even smaller heatsink than the Zotac motherboard and abnormally tall capacitors. The XFX GeForce 8200 and 8300 boards too were hot but they had slightly better cooling solutions with the heatpipes.

Performance

The performance of onboard graphics solutions for the AMD platform has typically been better than the ones for Intel platforms. For example in all the game benchmarks, the performance of the boards wasn't a lot different. In *Doom 3*, at a resolution of 640 x 480 and ultra quality, the frame rates recorded were between 46.6 and 60.5.

Of all the motherboards, the Jetway HA07-LF was the best overall performer. The AMD 790GX chipset is an even more powerful version of the famous 780G chipset. *Doom 3* will also run fairly smoothly at 1024 x 768 with quality set to maximum. The HA07 board recorded 35.4 fps at this setting. The next best performer is the other AMD 790GX powered board — the Biostar TA790 GX XE.

PCMark 05 didn't really show any big jump in numbers, so desktop performance was more or less the same — whether boards were priced at Rs 4,000 or at Rs 15,500. The graphics and the memory handling were the only two areas that affected the overall PCMark 05 score. The only real benchmark other than the games that are affected is the file transfer test. The XFX MI-A78S-8209 took 213 seconds for copying a 4 GB data file. Unlike the Intel based chipsets where the CPU score would drastically be affected in 3D Mark, the CPU scores for motherboards for the AMD platform were a lot more stable with both the NVIDIA boards as well as the AMD ones.

Bundled Goodies

All the motherboards for Intel and AMD processors come with the bare necessary items, there are some who go overboard. ASUS especially supplies addons and games with almost every motherboard. The ASUS M3N-HT Deluxe comes with a heatpipe for the memory that can be attached to it. It is only meant for two RAM sticks though. The Crosshair II Formula board is part of the Republic of Gamers range of boards from ASUS. A copy of *Company of Heroes - Opposing Fronts* is bundled along with it. An LED poster also comes with the package.

Value For Money

The ASUS Cross II Formula receives the award as the Best Performer of all the AMD boards. This is totally a hardcore overclocker board. It comes in a great package and everything about it exudes quality — everything from the BIOS to the quality of the board and the bundle supplied. All this comes at a price of Rs 15,450.

The most impressive value for money board has to be the Jetway HA07-LF with its unbeatable price



Jetway's HA 07 is cheap but is the best performer of all the boards for the AMD platform

of just Rs 6,750. A good open layout, great onboard performance, even better overclocking features and it also supports CrossFire. The HA06 is another similar feature-filled board based on the AMD 780G chipset with CrossFire but only a little slower than the HA07. The HA06 is priced at just Rs 5,500.

Another really good motherboard that comes at a fairly low price is the Biostar GeForce 8200 M2+ that sells for just Rs 3,900. The board doesn't come with CrossFire but it makes an excellent buy for someone who wants a really cheap motherboard for an entry level AMD processor. The Palit AA-780G is another cheap board available for Rs 200 less than the Biostar GeForce 8200 M2+. It's an equally compact board that is left behind a bit in the gaming performance.

The MSI DKA790GX Platinum is the next overall best board. This is yet another board powered by AMD's 790GX. At Rs 9,600, it's more affordable than the ASUS Crosshair Formula II and the ASUS M3N-HT Deluxe, performs almost as good as them, has a pretty good BIOS and a clean layout.

A similar conclusion like the boards for the Intel platform can be reached by just looking at the performance figures for the AMD platform boards. The performance that really matters to a lot of the people using computers today- everything to do with desktop applications like browsers, mail clients and office suites, there isn't a whole new world of performance waiting you while upgrading to a better motherboard. A motherboard is something that you look for as a base to build an entire system on. Features, reliability and the ability to upgrade and expand are the key - not performance. This is clearly seen in this test where we see boards priced at two times higher perform more or less the same way as their cheaper counterparts. A brief guide to what processor and motherboard you should go for lies ahead...

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Security is a matter of trust.



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- Anti Piracy
- Web Authentication
- Time Leasing



Biometrics *BioSentry*

- Time Attendance System
- Physical Access Control
- Logical Access Control (PC Logon)
- Data Security



Smart Card *SmartSentry*

- Time Attendance System
- Physical Access Control



Software & Other Applications

- Time Attendance Software
- Payroll Software
- Canteen Management
- Visitors Management
- Library Management, etc

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Time Attendance
Access Directory/ID Card
Access Control

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